

CHLOE H. CHOI

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RESEARCH SUMMARY

My research combines **computational fluid dynamics**, **scientific machine learning**, and **multi-fidelity uncertainty quantification** to develop **cardiovascular digital twins**. I develop end-to-end, data-to-prediction frameworks that enable efficient and reliable parameter estimation for clinical workflows.

EDUCATION

Stanford University

2023—present

Anticipated Degree: Ph.D. in Mechanical Engineering

California Institute of Technology (Caltech)

2019—2023

B.S. in Mechanical Engineering, Thesis: Towards a Reduced-Order Model of Burst-Wave Lithotripsy

RESEARCH APPOINTMENTS

Stanford University

Sept **2023**—Present

Yansouni Family Stanford Graduate Fellow — Adviser: Alison Marsden, Co-adviser: Gianluca Iaccarino

- Developing multi-fidelity parameter estimation methods for interventional planning of peripheral pulmonary artery stenosis (PPAS).
- First-year PhD rotations: Beverley McKeon (Fall 2023), Alison Marsden (Winter 2024), Todd Coleman (Spring 2024)

California Institute of Technology

Sept **2022**—June **2023**

Senior Thesis — Adviser: Tim Colonius

- Developed a two-dimensional data-driven reduced-order model of burst-wave lithotripsy to determine optimal frequencies to break kidney stones.

Massachusetts Institute of Technology

June **2022**—Aug **2022**

Summer Undergraduate Research Fellow — Adviser: Themis Sapsis

- Combined Bayesian optimization, extreme acquisition functions, and an ensemble of deep neural operators to predict rogue waves in high dimensions and jet acoustic bursts.

California Institute of Technology

June **2021**—Nov **2021**

Summer Undergraduate Research Fellow — Adviser: Tim Colonius

- Analyzed optimized eddy-viscosity models for coherent structures in the first and second azimuthal modes of turbulent jets and examined the physical significance of the approach to mechanisms present in the jet.

California Institute of Technology

June **2020**—Dec **2020**

Summer Undergraduate Research Fellow — Adviser: Julia Greer

- Investigated the fibrous properties of the *A. aurita* jellyfish mesoglea to characterize the viscoelastic response under flat punch indentation and to model radial symmetrization.

Massachusetts Institute of Technology

June **2018**—July **2018**

High School Intern — Adviser: Kerri Cahoy

- Modified a receiver assembly for MIT's Portable Telescope for Lasercom (PorTeL) to improve tracking performance. Modeled the orbit of MIT's Deformable Mirror (DeMi) mission.

PUBLICATIONS

Journal Publications

- J. Jilberto, **C. H. Choi**, J. Hwang, A. K. Reed, Z. Hu, W. Yang, & A. L. Marsden. "Digital Twin Frameworks for Treatment Planning in Congenital Heart Disease." Submitted, by invitation from *Current Treatment Options in Cardiovascular Medicine*.
- **C. H. Choi**, A. L. Marsden, & D. E. Schiavazzi. "A flow matching-based amortized inference framework for patient-specific cardiovascular models." [arXiv.org/abs/2603.19331](https://arxiv.org/abs/2603.19331). Under revision in *npj Cardiovascular Health*.
- **C. H. Choi**, A. Zanoni, D. E. Schiavazzi, & A. L. Marsden. "On the performance of multi-fidelity and reduced-dimensional neural emulators for inference of physiologic boundary conditions." *Computers in Biology and Medicine*, 200 (2026): 111389. doi: 10.1016/j.combiomed.2025.111389.

Book Chapters

- **C. H. Choi**, A. Zanoni, D. E. Schiavazzi, & A. L. Marsden. "A primer on uncertainty quantification for cardiovascular simulations." *Model Validation and Uncertainty Quantification in Biomechanics and Soft Tissues*, edited by G. Holzapfel, M. Rolf-Pissarczyk, & X. Y. Xu. In production.
- F. Schäfer, **C. H. Choi**, A. Zanoni, K. Menon, G. Geraci, A. L. Marsden, & D. E. Schiavazzi. "An introduction to multi-fidelity uncertainty propagation for applications in biomechanics." *Model Validation and Uncertainty Quantification in Biomechanics and Soft Tissues*, edited by G. Holzapfel, M. Rolf-Pissarczyk, & X. Y. Xu. In production.

CONFERENCE PARTICIPATION

- **C. Choi**, D. E. Schiavazzi, & A. L. Marsden. "Uncertainty-aware amortized inference of physiologic boundary conditions." *8th Annual Cardiac Imaging, Mechanics, and Modeling (CIM2)*, Stanford University, Palo Alto, CA, August 2026.

- **C. H. Choi**, D. E. Schiavazzi, & A. L. Marsden. “Uncertainty-aware amortized inference of boundary conditions for vascular anatomies with stenosis.” *9th International Conference on Computational & Mathematical Biomedical Engineering (CMBE2026)*, Kobe, Japan, June 2026.
- **C. Choi**, D. E. Schiavazzi, & A. L. Marsden. “Amortized inference of boundary conditions.” *SIAM Conference on Uncertainty Quantification (SIAM UQ26)*, Minneapolis, MN, March 2026.
- **C. Choi**, A. Zanoni, D. E. Schiavazzi, & A. L. Marsden. “Inference of boundary conditions using multi-fidelity and reduced-dimensional neural emulators.” *7th Annual Cardiac Imaging, Mechanics, and Modeling (CIM2)*, Stanford University, Palo Alto, CA, August 2025.
- **C. Choi**, A. Zanoni, D.E. Schiavazzi, & A.L. Marsden. “Learning the discrepancy between low- and high-fidelity models to improve parameter estimation for cardiovascular simulations.” *18th Annual US National Congress on Computational Mechanics (USNCCM18)*, Chicago, IL, July 2025.
- **C. Choi**, A. Zanoni, D. E. Schiavazzi, & A. L. Marsden. “Learning the discrepancy between low- and high-fidelity models to improve parameter estimation for cardiovascular simulations.” *ICERM Uncertainty Quantification for Mathematical Biology Workshop*, Providence, RI, May 2025.
- **C. Choi**, A. Zanoni, D.E. Schiavazzi, & A.L. Marsden. “Learning the discrepancy between low- and high-fidelity models to improve parameter estimation for cardiovascular simulations.” *SIAM Conference on Computational Science and Engineering (SIAM CSE25)*, Fort Worth, TX, March 2025.
- T.E. Colonius, J.S. Spratt, A. Chandra, **C. Choi**, & C. Li. “PPG Update - Caltech.” *Annual Program Project Grant (PPG) Meeting*, Seattle, WA, August 2023. (T Colonius presented.)
- **C.H. Choi**, E.M. Pickering, & T.E. Colonius. “Optimized eddy-viscosity models for coherent structures in turbulent jets.” *74th Annual Meeting of the APS Division of Fluid Dynamics (DFD)*, Phoenix, AZ, November 2021.

INVITED LECTURES AND WORKSHOPS

- **Tutorial**, “Boundary condition tuning with svSuperEstimator,” Cardiovascular Biomechanics Computation Laboratory, Stanford University, Jan 2025.

GRANTS AND AWARDS

NSF ACCESS Discover Grant (MCH260150), *ACCESS Allocations* 2026–2027

Awarded 1.5M ACCESS credits to pursue independent dissertation project with high-performance computing resources.

Rising Stars in Mechanical Engineering, *Stanford* 2026

Selective workshop consisting of 30 participants based on potential for an academic career in mechanical engineering.

CMBE Travel Grant Award, *CMBE2026, Kobe, Japan* 2026

\$500 travel fund awarded based on the quality, originality, and impact of the accepted abstract as well as the relevance to the conference themes.

CVI Travel Award, *Stanford Cardiovascular Institute (CVI)* 2026

\$750 travel fund awarded by application process.

AHA Predoctoral Fellowship (26PRE1550972), *American Heart Association (AHA)* 2026–2027

\$70,676 fellowship awarded to outstanding predoctoral students. The fellowship provides a stipend for two years and opportunities for professional development and networking with other AHA fellows and alumni.

Travel fund for ICERM UQ for Mathematical Biology, *ICERM Brown* 2025

\$800 travel fund awarded by application process.

Yansouni Family Stanford Graduate Fellowship in Science & Engineering (SGF), *Stanford* 2023–2026

Three-year fellowship awarded to outstanding PhD students at Stanford University. The fellowship provides full tuition and a stipend, as well as opportunities for professional development and networking with other SGF fellows and alumni.

Chryssostomidis/Ocean Engineering Graduate Departmental Fellowship, *MIT* 2023

(Declined) Full tuition, stipend, insurance coverage for outstanding incoming PhD students by selective nomination.

Francis Robbins Upton Fellowship in Engineering, *Princeton* 2023

(Declined) Highest honor for incoming outstanding PhD students in engineering.

Summer Undergraduate Research Fellowship (SURF), *Caltech* 2021, 2022

George W. Housner Student Discovery Fund, *Caltech* 2021

The Housner Fund, established through a gift from the late George W. Housner, Braun Professor Emeritus in Engineering, supports scholarly activities that enrich the undergraduate academic experience. Awarded \$1000.

Kiyo and Eiko Tomiyasu Scholarship, *Caltech* 2020

Named Summer Undergraduate Research Fellowship (SURF) program fellows benefit from funds generously contributed by donors to support the SURF program.

PROFESSIONAL MEMBERSHIPS

American Physical Society (APS) 2021—present

Society for Industrial and Applied Mathematics (SIAM) 2025—present

American Heart Association (AHA) 2025—present

Stanford Cardiovascular Institute (CVI) 2026—present

PROFESSIONAL SERVICE

Journal peer review:

- Annals of Biomedical Engineering (co-reviewed with Prof. Marsden) 2026
- Cardiovascular Engineering and Technology 2026
- Computer Methods in Applied Mechanics and Engineering 2026

Judge/panel:

- Poster judge (by invitation), Stanford Cardiovascular Research Symposium 2026

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant, Computational Modeling in the Cardiovascular System (ME285), Stanford Fall **2026**
(Upcoming) Hosted weekly office hours and created/graded homework and solutions. *Topics covered:* incompressible computational fluid dynamics, lumped parameter models, patient specific modeling with finite elements, uncertainty quantification.

Undergraduate Project Co-Supervision, Stanford Spring **2026**—Summer **2026**
Cardiovascular Institute (CVI) Summer Research Fellow, Project: “Virtual treatment planning for peripheral pulmonary arterial stenosis.” Won Poster Award at Cardiovascular Research Symposium (CVRS).

Visiting PhD Researcher Project Supervision, Stanford Spring **2026**—Present
“Uncertainty-aware risk stratification of aneurysms caused by Kawasaki disease”

Undergraduate Junior Honors Thesis Co-Supervision, Stanford Fall **2024**—Spring **2025**
CS Junior Honors Thesis: “Personalized Digital Twins for Coronary Artery Disease Treatment and Reduced-Order Modeling”

Undergraduate Project Co-Supervision, Stanford Summer **2024**
BioE REU Fellow: “Interactive Web Apps for Designing and Visualizing Zero-Dimensional Hemodynamic Models”

Undergraduate Project Supervision, Caltech Summer **2023**
WAVE Fellow: “Optimize Burst Wave Lithotripsy: MFC and Strain Energy”

Teaching Assistant, Dimensional and Data Analyses in Engineering (ME40), Caltech Fall **2021**
Hosted weekly office hours, graded homework, and created solutions. *Topics covered:* Buckingham Pi theorem, random variables, linear regression/least-squares, principal components, spectral/Fourier analysis

LEADERSHIP AND OUTREACH

Committee on Research, University Committee Representative, Stanford Fall **2026**—Present
(Upcoming) Advised on institutional research policy, sponsored funding procedures, and faculty research governance, and attended committee meetings. Selected via application and interview.

Stanford seeME, Stanford Spring **2026**
Mechanical engineering outreach program for local middle- and high-school students from underrepresented backgrounds. Developed and taught a class to design a physics-based stunt jump track for a Hot Wheels car in seeME’s annual event.

Stories of Women in Fluids (SOWIF) Initiative Winter **2026**—Present
Voluntary, not-for-profit initiative to inspire women to mentor and motivate girls and fellow women towards successful fluid dynamics careers through a dynamic written platform.

Peer Advocate Program, Caltech Spring **2022**—Spring **2023**
Developed monthly community-building activities for residents and raised mental health awareness to enhance the residential experiences of students. Selected via application.

Student-Faculty Programs (SFP) Ambassador, Caltech Fall **2021**—Winter **2022**
Organized and facilitated informational sessions for residents while serving as a peer advisor. Selected via nomination.

Academics and Research Committee (ARC) Representative, Caltech Fall **2019**—Spring **2021**
Academic branch of student government; selected via application and interview. Organized student-faculty lunches, voiced course concerns to improve academic life, and started ARC’s Facebook page. Led discussions with faculty on improving the mechanical engineering curriculum based on student survey results at the 2021 Student-Faculty Conference.